

Android Studio interface showing the development of an Android application named **Experiment12\_SensorDemo** on a Samsung SM-G781B device.

**Project Structure:**

- Experiment1\_HelloWorld
  - manifests
    - AndroidManifest.xml
  - kotlin+java
    - com.example.helloworld
      - MainActivity.kt
  - res (generated)
- Experiment6\_QuoteApp
- Experiment7\_SMSAutoReply
- Experiment8\_BluetoothScanner
- Experiment9\_BluetoothChat
- Experiment10\_WiFiSignal
- Experiment11\_GPSMaps
- Experiment12\_SensorDemo
  - manifests
  - kotlin+java
    - com.example.sensordemo
      - MainActivity.kt
  - res (generated)
- Experiment13\_FirebaseChat
- Experiment14\_MLKitFaceDetection
- Gradle Scripts

**MainActivity.kt Code:**

```
61 class MainActivity : AppCompatActivity() {
62     override fun onCreate(savedInstanceState: Bundle?) {
63         setContentView(R.layout.activity_main)
64         SensorDemoTheme {
65             // ...
66         }
67     }
68 }
69
70 1 Usage
71 private fun startSensors() {
72     accelerometer?.also { sensorManager?.registerList
73     gyroscope?.also { sensorManager?.registerListene
74     isListening = true
75 }
76
77 2 Usages
78 private fun stopSensors() {
79     sensorManager?.unregisterListener(this)
80     isListening = false
81 }
82
83 override fun onSensorChanged(event: SensorEvent?) {
84     event?.let {
85         when (it.sensor.type) {
86             // ...
87         }
88     }
89 }
```

**Build Output:**

Build Experiments: finished At 25/08/26, 3:24 | 1 sec, 870 ms

34 actionable tasks: 6 executed, 28 up-to-date

Configuration cache entry stored.

[Build Analyzer](#) results available

**Sensor Demo App Preview:**

Sensor Demo

Start Stop

**Accelerometer (m/s²)**

| Axis      | Value |
|-----------|-------|
| X         | -1.88 |
| Y         | 0.88  |
| Z         | 10.96 |
| Magnitude | 11.15 |

**Gyroscope (rad/s)**

| Axis      | Value |
|-----------|-------|
| X         | -0.50 |
| Y         | 0.09  |
| Z         | 0.91  |
| Magnitude | 1.04  |

46%